Preimage cryptanalysis of the baseline mixed output: efforts and conclusions

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1 Objective and setup

The target is the declared mixed output of the baseline / as-is run of the TDP pipeline: a reversible circuit **F** on 256 wires with **666,745 gates** that computes a 64-bit permutation **C** on the honest input slice. The honest input places x on wires 0–63 and zeros elsewhere; the payload $C(x)$ appears on wires 64–127 at the output.

Goal. Find an input x with $C(x) = 0^{64}$ (a preimage of zero), using only the declared output **F** and the attacker's own rewrites of it. The identity of the internal C values exposed by the earlier census is *not* used — it was derived with knowledge of C and is therefore off-limits. C itself appears only in clearly labelled diagnostics that measure what an attack could have seen; it never drives an attack. Every candidate x is verified against the real circuit.

Three equivalent forms of the same C were available and used as targets:

circuit	gates	origin
F	666,745	declared baseline/as-is mixed output
plateau	355,990	C-blind ANF resynthesis of F (30 rounds)
short-279k	279,263	fcompress shortening sweep (`fc4_c5_n64`), verified equivalent

A preimage found on any of them is the same x, verifiable on all.

2 Attacks attempted

2.1 Direct SAT inversion

Each circuit was encoded into CNF (one variable per initial wire and per gate birth; Toffoli gates as an AND auxiliary plus a native XOR clause for CryptoMiniSat), with the non-input wires fixed to 0 at entry and the payload fixed to 0^{64} at exit. Instances on F and on the plateau ran with 8 threads each. **Outcome:** no solution; both baselines ran more than 30 hours without returning.

2.2 Structural reconstruction (vocabulary walks)

Build the vocabulary of every function (values on 8,192 sampled inputs, up to complement) born on any wire of the derived circuits, then walk `g57` gates forward from the inputs and backward from the payload, accepting a gate whenever the implied value is present in the vocabulary. A complete chain would reconstruct C, hence x. **Outcome:** no chain. The vocabulary covers 684 of C's 1,000

gate births with seven circuits and 735 with all 65 fair-game circuits, but mere membership does not identify a genuine C gate: the C-blind resyntheses and stage-3 remixes contain many non-C values of exactly the tested form. Forward walks were spurious from the first gate; backward walks were genuine only with the smallest vocabulary and stalled after 20–40 gates. A lineage rule and a degree-2 span filter reduced false matches but did not make the walk reliable.

2.3 The replication landmark

The raw sandwich computes $C(x)$ in the middle and copies it, so an unmixed circuit has an interior cut where two disjoint 64-wire bundles carry the same functions; constraining that bundle to zero would isolate C in the first half. We tested, C-blind, whether this survives, at three levels of generality.

- **Verbatim / up to NOT — absent.** Over every cut of F and its three rewrites, the number of functions carried by two or more wires peaks at only 15–21 and only at the very start (depth 0.005–0.009, the input-copy step); it is zero through the whole interior. F's 64 output functions are never co-present at an interior cut (all 64 only at the final gate). Stage-6 packing and ANF resynthesis rebuild the circuit toward the output, destroying the clean mid-circuit copy.
- **Affine (degree-1) — weak.** F peaks at only 10 of 64 output bits in the affine span of the live wires at depth 0.485; the two remixes at 5. Insufficient to invert.
- **Degree-2 — the finding (Section 3).**

2.4 Exploiting the degree-2 structure

Three sound ways to turn the degree-2 midpoint relations into an inversion were built and run; all add constraints logically implied by payload = 0 and all are self-verifying.

- **Two-circuit linked SAT.** Encode the plateau and F over a shared 64-variable input, both payloads zero, plus clauses equating internal variables whose functions coincide on samples. The two equivalent circuits share **42,771** identical internal functions, so a value forced cheaply in the small circuit propagates into the large one. Variants with 6,000 and 30,000 links ($\sim$1.7M variables).
- **Hint-augmented SAT.** The full plateau with payload = 0, plus the sparse degree-2 relations imposed at one cut (50 hints) and at two cuts (94 hints, covering 59 of 64 output bits) as redundant implied constraints.
- **Prefix-only SAT.** Encode only the prefix up to the degree-2 cut, impose the relations there, and enumerate the prefix's satisfying inputs, checking each against the full circuit. On the plateau, 59 constraints across two cuts pin the input to about 2^5 candidates over 178k of the 356k gates. On short-279k the earlier midpoint gives a shorter prefix (112k gates) but only 40 constraints.

Outcome: no candidate or preimage from any variant. The prefix solve — the best-shaped, smallest instance — produced zero satisfying inputs after 16+ hours on the 279k circuit and 24+ hours on the plateau; the linked and hint variants likewise returned nothing.

3 Key finding: the degree-2 midpoint replica

The replication landmark survives the pipeline **not verbatim and not affinely, but up to degree 2**. At each cut we measured how many of the 64 output functions are GF(2) polynomials of degree at most 2 in the live wires, over enough samples (36,864) that the ∼33,000 degree-2 monomials stay below the sample count (membership false-positive $\approx 2^{-4000}$, so the counts are not coincidental). The count rises sharply toward the middle and peaks per circuit:

circuit	peak deg-2 bits	depth	profile at 0.40/0.45/0.48/0.50
plateau (356k)	50 of 64	0.45	13 / 50 / 30 / 44
F (666k)	36 of 64	0.48	3 / 20 / 36 / 19
short-279k	40 of 64	0.40	40 / 37 / 26 / 29
remix (1.38M)	20 of 64	0.48	2 / 12 / 20 / 12
remix2 (1.11M)	16 of 64	0.50	3 / 8 / 8 / 16

Three properties make this a genuine, usable weakness rather than an artifact:

- **Sparse.** Despite the near-full-rank degree-2 span, the representations are compact: on the plateau at depth 0.45 the 50 relations use a median of 10 monomials (max 50), and four output bits (17, 31, 41, 57) equal a single interior wire verbatim. On the 279k circuit the median is 6.
- **Nested and growing.** On the plateau at depth 0.45, by minimal replica: 4 bits are a bare wire, 5 are affine, 50 are degree-2, 14 have none. The classes nest, so higher degree exposes genuinely new bits — the affine-to-quadratic jump is almost entirely new (45 of the 50 need a product).
- **Complementary across circuits.** The plateau and F expose largely different bits (overlap 29). Two plateau cuts cover 59 of 64; adding F's cut reaches **62 of 64**. Almost every output bit is a compact degree-2 function of interior wires at some midpoint cut.

An incidental observation: each step of ANF resynthesis *increased* the degree-2 exposure (666k → 36, 356k → 50), while the fcompress-shortened 279k circuit peaks lower (40) and earlier (depth 0.40) — so the low-degree leak comes specifically from ANF resynthesis, not from compression in general.

4 Conclusions

1. **No preimage was found.** Across three equivalent sizes (666k, 356k, 279k) and five sound SAT formulations (full, two-circuit linked, degree-2 hinted at one and two cuts, half-circuit with the bundle constrained), no solver produced x with $C(x) = 0^{64}$. The best-shaped instance returned no satisfying assignment at all.
2. **The structural leak is real and nearly complete.** The sandwich's mid-circuit replication of C survives at degree 2: around the midpoint, almost every output bit (62 of 64 across two circuits) is a sparse degree-2 function of the interior wires — a concrete, quantified weakness the verbatim and affine scans missed.
3. **The leak does not translate into a feasible inversion by these means.** The obstacle is the solver, not the structure: inverting a 100k–180k-gate reversible circuit is intractable for CryptoMiniSat here even with the degree-2 relations pinning the input and the uncompute half removed. Shrinking from 666k to 279k did not cross the threshold.

4. **Interpretation.** The degree-2 replica is a property of the *representation* — resynthesis expresses C's output as low-degree functions of its own intermediates — not a shortcut to x. Exploiting it still requires solving for an input consistent with the prefix, the hard part. A different engine (algebraic / Gröbner over the sparse quadratic system, or a dedicated reversible-circuit inverter) is the next thing to try; standard SAT is not enough.

5 Tools and artifacts

All tools are on the server under `~/threemask_cmp/py6/`; results under `~/threemask_cmp/run/preimage/`. Direct SAT: `sat_invert.py`. Vocabulary walks: `vocab.py`, `blind_walk3.py`, `vocab_diag*.py`. Replication scans: `payload_cut.py`, `replication_scan.py`, `affine_replica_scan.py`, `degree2_replica_scan.` Combinations: `combo_extract.py`, `classify_replicas.py`. Exploits: `combined_link_sat.py`, `plateau_hint_sat.py`, `plateau_hint_sat2.py`, `prefix_quad_sat.py`. The full chronological log is in `docs/notes/stage2_stage6_campaign_20261021.md`.